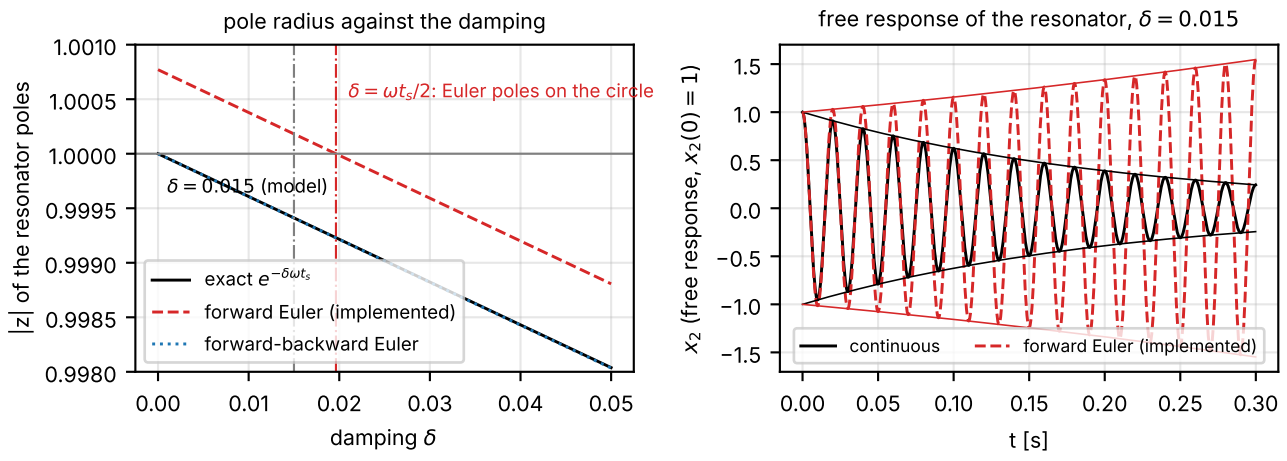
pole radius against the damping
1.0010
1.0005
1.0000
0.9995
0.9990
0.9985
0.9980
|z| of the resonator poles
δ = ωt_s/2: Euler poles on the circle
δ = 0.015 (model)
exact e^{−δωt_s}
forward Euler (implemented)
forward-backward Euler
0.00
0.01
0.02
0.03
0.04
0.05
damping δ
free response of the resonator, δ = 0.015
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
x_2 (free response, x_2(0) = 1)
continuous
forward Euler (implemented)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
t [s]